

Modern Physical Organic Chemistry

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Eric V. Anslyn

UNIVERSITY OF TEXAS, AUSTIN

Dennis A. Dougherty

CALIFORNIA INSTITUTE OF TECHNOLOGY



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Abbreviated Contents

PART I: Molecular Structure and Thermodynamics

- CHAPTER 1. Introduction to Structure and Models of Bonding 3
2. Strain and Stability 65
3. Solutions and Non-Covalent Binding Forces 145
4. Molecular Recognition and Supramolecular Chemistry 207
5. Acid-Base Chemistry 259
6. Stereochemistry 297
-

PART II: Reactivity, Kinetics, and Mechanisms

- CHAPTER 7. Energy Surfaces and Kinetic Analyses 355
8. Experiments Related to Thermodynamics and Kinetics 421
9. Catalysis 489
10. Organic Reaction Mechanisms, Part 1:
Reactions Involving Additions and/or Eliminations 537
11. Organic Reaction Mechanisms, Part 2:
Substitutions at Aliphatic Centers and Thermal
Isomerizations/Rearrangements 627
12. Organotransition Metal Reaction Mechanisms and Catalysis 705
Intent and Purpose 705
13. Organic Polymer and Materials Chemistry 753
-

PART III: Electronic Structure: Theory and Applications

- CHAPTER 14. Advanced Concepts in Electronic Structure Theory 807
15. Thermal Pericyclic Reactions 877
16. Photochemistry
17. Electronic Organic Materials 1001
-

- APPENDIX 1. Conversion Factors and Other Useful Data 0000
2. Electrostatic Potential Surfaces for Representative Organic Molecules 0000
3. Group Orbitals of Common Functional Groups:
Representative Examples Using Simple Molecules 0000
4. The Organic Structures of Biology 0000
5. Pushing Electrons 0000
6. Reaction Mechanism Nomenclature 0000

List of Highlights	00
Preface	00
Acknowledgments	00
A Note to the Instructor	00

PART I

MOLECULAR STRUCTURE AND THERMODYNAMICS

CHAPTER 1: Introduction to Structure and Models of Bonding 3

Intent and Purpose 3

1.1 A Review of Basic Bonding Concepts	4
1.1.1 Quantum Numbers and Atomic Orbitals	4
1.1.2 Electron Configurations and Electronic Diagrams	5
1.1.3 Lewis Structures	6
1.1.4 Formal Charge	6
1.1.5 VSEPR	7
1.1.6 Hybridization	8
1.1.7 A Hybrid Valence Bond/Molecular Orbital Model of Bonding	10
<i>Creating Localized σ and π Bonds</i>	11
1.1.8 Polar Covalent Bonding	12
<i>Electronegativity</i>	12
<i>Electrostatic Potential Surfaces</i>	14
<i>Inductive Effects</i>	15
<i>Group Electronegativities</i>	
<i>Hybridization Effects</i>	
1.1.9 Bond Dipoles, Molecular Dipoles, and Quadrupoles	17
<i>Bond Dipoles</i>	17
<i>Molecular Dipole Moments</i>	18
<i>Molecular Quadrupole Moments</i>	19
1.1.10 Resonance	20
1.1.11 Bond Lengths	22
1.1.12 Polarizability	24
1.1.13 Summary of Concepts Used for the Simplest Model of Bonding in Organic Structures	26

1.2 A More Modern Theory of Organic Bonding 26

1.2.1 Molecular Orbital Theory	27
1.2.2 A Method for QMOT	28
1.2.3 Methyl in Detail	29
<i>Planar Methyl</i>	29
<i>The Walsh Diagram: Pyramidal Methyl</i>	31
<i>"Group Orbitals" for Pyramidal Methyl</i>	32
<i>Putting the Electrons In—The MH_3 System</i>	33
1.2.4 The CH_2 Group in Detail	33
<i>The Walsh Diagram and Group Orbitals</i>	33
<i>Putting the Electrons In—The MH_2 System</i>	33

1.3 Orbital Mixing—Building Larger Molecules 35

1.3.1 Using Group Orbitals to Make Ethane	36
---	----

1.3.2 Using Group Orbitals to Make Ethylene	38
1.3.3 The Effects of Heteroatoms—Formaldehyde	40
1.3.4 Making More Complex Alkanes	43
1.3.5 Three More Examples of Building Larger Molecules from Group Orbitals	43
<i>Propene</i>	43
<i>Methyl Chloride</i>	45
<i>Butadiene</i>	46
1.3.6 Group Orbitals of Representative π Systems: Benzene, Benzyl, and Allyl	46
1.3.7 Understanding Common Functional Groups as Perturbations of Allyl	49
1.3.8 The Three Center–Two Electron Bond	50
1.3.9 Summary of the Concepts Involved in Our Second Model of Bonding	51

1.4 Bonding and Structures of Reactive Intermediates 52

1.4.1 Carbocations	52
<i>Carbenium Ions</i>	53
<i>Interplay with Carbonium Ions</i>	54
<i>Carbonium Ions</i>	55
1.4.2 Carbanions	56
1.4.3 Radicals	57
1.4.4 Carbenes	58

1.5 A Very Quick Look at Organometallic and Inorganic Bonding 59

Summary and Outlook 61

EXERCISES 62

FURTHER READING 64

CHAPTER 2: Strain and Stability 65

Intent and Purpose 65

2.1 Thermochemistry of Stable Molecules 66

2.1.1 The Concepts of Internal Strain and Relative Stability	66
2.1.2 Types of Energy	68
<i>Gibbs Free Energy</i>	68
<i>Enthalpy</i>	69
<i>Entropy</i>	70
2.1.3 Bond Dissociation Energies	70
<i>Using BDEs to Predict Exothermicity and Endothermicity</i>	72
2.1.4 An Introduction to Potential Functions and Surfaces—Bond Stretches	73
<i>Infrared Spectroscopy</i>	77
2.1.5 Heats of Formation and Combustion	77
2.1.6 The Group Increment Method	79
2.1.7 Strain Energy	82

2.2 Thermochemistry of Reactive Intermediates 82

2.2.1 Stability vs. Persistence 82

2.2.2 Radicals 83

BDEs as a Measure of Stability 83*Radical Persistence* 84*Group Increments for Radicals* 86

2.2.3 Carbocations 87

Hydride Ion Affinities as a Measure of Stability 87*Lifetimes of Carbocations* 90

2.2.4 Carbanions 91

2.2.5 Summary 91

**2.3 Relationships Between Structure and Energetics—
Basic Conformational Analysis 92**

2.3.1 Acyclic Systems—Torsional Potential Surfaces 92

Ethane 92*Butane—The Gauche Interaction* 95*Barrier Height* 97*Barrier Foldedness* 97*Tetraalkylethanes* 98*The g+g–Pentane Interaction* 99*Allylic ($A^{1,3}$) Strain* 100

2.3.2 Basic Cyclic Systems 100

Cyclopropane 100*Cyclobutane* 100*Cyclopentane* 101*Cyclohexane* 102*Larger Rings—Transannular Effects* 107*Group Increment Corrections for Ring Systems* 109*Ring Torsional Modes* 109*Bicyclic Ring Systems* 110*Cycloalkenes and Bredt's Rule* 110*Summary of Conformational Analysis and**Its Connection to Strain* 112**2.4 Electronic Effects 112**2.4.1 Interactions Involving π Systems 112*Substitution on Alkenes* 112*Conformations of Substituted Alkenes* 113*Conjugation* 115*Aromaticity* 116*Antiaromaticity, An Unusual Destabilizing Effect* 117*NMR Chemical Shifts* 118*Polycyclic Aromatic Hydrocarbons* 119*Large Annulenes* 119

2.4.2 Effects of Multiple Heteroatoms 120

Bond Length Effects 120*Orbital Effects* 120**2.5 Highly-Strained Molecules 124**

2.5.1 Long Bonds and Large Angles 124

2.5.2 Small Rings 125

2.5.3 Very Large Rotation Barriers 127

2.6 Molecular Mechanics 128

2.6.1 The Molecular Mechanics Model 129

Bond Stretching 129*Angle Bending* 130*Torsion* 130*Nonbonded Interactions* 130*Cross Terms* 131*Electrostatic Interactions* 131*Hydrogen Bonding* 131*The Parameterization* 132*Heat of Formation and Strain Energy* 1322.6.2 General Comments on the Molecular
Mechanics Method 1332.6.3 Molecular Mechanics on Biomolecules and
Unnatural Polymers—"Modeling" 135

2.6.4 Molecular Mechanics Studies of Reactions 136

Summary and Outlook 137**EXERCISES 138****FURTHER READING 143**

**CHAPTER 3: Solutions and Non-Covalent
Binding Forces 145****Intent and Purpose 145****3.1 Solvent and Solution Properties 145**

3.1.1 Nature Abhors a Vacuum 146

3.1.2 Solvent Scales 146

Dielectric Constant 147*Other Solvent Scales* 148*Heat of Vaporization* 150*Surface Tension and Wetting* 150*Water* 151

3.1.3 Solubility 153

General Overview 153*Shape* 154*Using the "Like-Dissolves-Like" Paradigm* 154

3.1.4 Solute Mobility 155

Diffusion 155*Fick's Law of Diffusion* 156*Correlation Times* 156

3.1.5 The Thermodynamics of Solutions 160

Chemical Potential 158*The Thermodynamics of Reactions* 160*Calculating ΔH° and ΔS°* 162**3.2 Binding Forces 162**

3.2.1 Ion Pairing Interactions 163

Salt Bridges 164

3.2.2 Electrostatic Interactions Involving Dipoles 165

Ion–Dipole Interactions 165*A Simple Model of Ionic Solvation —**The Born Equation* 166*Dipole–Dipole Interactions* 168

3.2.3 Hydrogen Bonding 168

Geometries 169*Strengths of Normal Hydrogen Bonds* 171*i. Solvation Effects* 171*ii. Electronegativity Effects* 172*iii. Resonance Assisted Hydrogen Bonds* 173*iv. Polarization Enhanced Hydrogen Bonds* 174*v. Secondary Interactions in Hydrogen**Bonding Systems* 175

vi. Cooperativity in Hydrogen Bonds	175
Vibrational Properties of Hydrogen Bonds	176
Short–Strong Hydrogen Bonds	177
3.2.4 π Effects	180
Cation– π Interactions	181
Polar– π Interactions	183
Aromatic–Aromatic Interactions (π Stacking)	184
The Arene–Perfluoroarene Interaction	184
π Donor–Acceptor Interactions	186
3.2.5 Induced-Dipole Interactions	186
Ion–Induced-Dipole Interactions	187
Dipole–Induced-Dipole Interactions	187
Induced-Dipole–Induced-Dipole Interactions	188
Summarizing Monopole, Dipole, and Induced-Dipole Binding Forces	188
3.2.6 The Hydrophobic Effect	189
Aggregation of Organics	189
The Origin of the Hydrophobic Effect	192
3.3 Computational Modeling of Solvation	194
3.3.1 Continuum Solvation Models	196
3.3.2 Explicit Solvation Models	197
3.3.3 Monte Carlo (MC) Methods	198
3.3.4 Molecular Dynamics (MD)	199
3.3.5 Statistical Perturbation Theory / Free Energy Perturbation	200
Summary and Outlook	201
EXERCISES	202
FURTHER READING	204

CHAPTER 4: Molecular Recognition and Supramolecular Chemistry 207

Intent and Purpose 207

4.1 Thermodynamic Analyses of Binding Phenomena	207
4.1.1 General Thermodynamics of Binding	208
The Relevance of the Standard State	210
The Influence of a Change in Heat Capacity	212
Cooperativity	213
Enthalpy–Entropy Compensation	216
4.1.2 The Binding Isotherm	216
4.1.3 Experimental Methods	219
UV/Vis or Fluorescence Methods	220
NMR Methods	220
Isothermal Calorimetry	221
4.2 Molecular Recognition	222
4.2.1 Complementarity and Preorganization	224
Crowns, Cryptands, and Spherands—Molecular Recognition with a Large Ion–Dipole Component	224
Tweezers and Clefts	228
4.2.2 Molecular Recognition with a Large Ion Pairing Component	228
4.2.3 Molecular Recognition with a Large Hydrogen Bonding Component	230
Representative Structures	230

Molecular Recognition via Hydrogen Bonding in Water	232
4.2.4 Molecular Recognition with a Large Hydrophobic Component	234
Cyclodextrins	234
Cyclophanes	234
A Summary of the Hydrophobic Component of Molecular Recognition in Water	238
4.2.5 Molecular Recognition with a Large π Component	239
Cation– π Interactions	239
Polar– π and Related Effects	241
4.2.6 Summary	241
4.3 Supramolecular Chemistry	243
4.3.1 Supramolecular Assembly of Complex Architectures	244
Self-Assembly via Coordination Compounds	244
Self-Assembly via Hydrogen Bonding	245
4.3.2 Novel Supramolecular Architectures—Catenanes, Rotaxanes, and Knots	246
Nanotechnology	248
4.3.3 Container Compounds—Molecules within Molecules	249

Summary and Outlook 252

EXERCISES	253
FURTHER READING	256

CHAPTER 5: Acid–Base Chemistry 259

Intent and Purpose 259

5.1 Brønsted Acid–Base Chemistry	259
5.2 Aqueous Solutions	261
5.2.1 pK_a	261
5.2.2 pH	262
5.2.3 The Leveling Effect	264
5.2.4 Activity vs. Concentration	266
5.2.5 Acidity Functions: Acidity Scales for Highly Concentrated Acidic Solutions	266
5.2.6 Super Acids	270
5.3 Nonaqueous Systems	271
5.3.1 pK_a Shifts at Enzyme Active Sites	273
5.3.2 Solution Phase vs. Gas Phase	273
5.4 Predicting Acid Strength in Solution	276
5.4.1 Methods Used to Measure Weak Acid Strength	276
5.4.2 Two Guiding Principles for Predicting Relative Acidities	277
5.4.3 Electronegativity and Induction	278
5.4.4 Resonance	278
5.4.5 Bond Strengths	283
5.4.6 Electrostatic Effects	283
5.4.7 Hybridization	283

- 5.4.8 Aromaticity 284
- 5.4.9 Solvation 284
- 5.4.10 Cationic Organic Structures 285

5.5 Acids and Bases of Biological Interest 285

5.6 Lewis Acids/Bases and Electrophiles/ Nucleophiles 288

- 5.6.1 The Concept of Hard and Soft Acids and Bases, General Lessons for Lewis Acid–Base Interactions, and Relative Nucleophilicity and Electrophilicity 289

Summary and Outlook 292

EXERCISES 292

FURTHER READING 294

CHAPTER 6: Stereochemistry 297

Intent and Purpose 297

6.1 Stereogenicity and Stereoisomerism 297

- 6.1.1 Basic Concepts and Terminology 298
 - Classic Terminology* 299
 - More Modern Terminology* 301
- 6.1.2 Stereochemical Descriptors 303
 - R,S System* 304
 - E,Z System* 304
 - D and L* 304
 - Erythro and Threo* 305
 - Helical Descriptors—M and P* 305
 - Ent and Epi* 306
 - Using Descriptors to Compare Structures* 306
- 6.1.3 Distinguishing Enantiomers 306
 - Optical Activity and Chirality* 309
 - Why is Plane Polarized Light Rotated by a Chiral Medium?* 309
 - Circular Dichroism* 310
 - X-Ray Crystallography* 310

6.2 Symmetry and Stereochemistry 311

- 6.2.1 Basic Symmetry Operations 311
- 6.2.2 Chirality and Symmetry 311
- 6.2.3 Symmetry Arguments 313
- 6.2.4 Focusing on Carbon 314

6.3 Topicity Relationships 315

- 6.3.1 Homotopic, Enantiotopic, and Diastereotopic 315
- 6.3.2 Topicity Descriptors—Pro-R/Pro-S and Re/Si 316
- 6.3.3 Chirotopicity 317

6.4 Reaction Stereochemistry: Stereoselectivity and Stereospecificity 317

- 6.4.1 Simple Guidelines for Reaction Stereochemistry 317
- 6.4.2 Stereospecific and Stereoselective Reactions 319

6.5 Symmetry and Time Scale 322

6.6 Topological and Supramolecular Stereochemistry 324

- 6.6.1 Loops and Knots 325
- 6.6.2 Topological Chirality 326

- 6.6.3 Nonplanar Graphs 326
- 6.6.4 Achievements in Topological and Supramolecular Stereochemistry 327

6.7 Stereochemical Issues in Polymer Chemistry 331

6.8 Stereochemical Issues in Chemical Biology 333

- 6.8.1 The Linkages of Proteins, Nucleic Acids, and Polysaccharides 333
 - Proteins* 333
 - Nucleic Acids* 334
 - Polysaccharides* 334
- 6.8.2 Helicity 336
 - Synthetic Helical Polymers* 337
- 6.8.3 The Origin of Chirality in Nature 339

6.9 Stereochemical Terminology 340

Summary and Outlook 340

EXERCISES 344

FURTHER READING 350

PART II

Reactivity, Kinetics, and Mechanisms

CHAPTER 7: Energy Surfaces and Kinetic Analyses 355

Intent and Purpose 355

7.1 Energy Surfaces and Related Concepts 356

- 7.1.1 Energy Surfaces 357
- 7.1.2 Reaction Coordinate Diagrams 359
- 7.1.3 What is the Nature of the Activated Complex/Transition State? 362
- 7.1.4 Rates and Rate Constants 363
- 7.1.5 Reaction Order and Rate Laws 364
- 7.2 Transition State Theory (TST) and Related Topics 365
 - 7.2.1 The Mathematics of Transition State Theory 365
 - 7.2.2 Relationship to the Arrhenius Rate Law 367
 - 7.2.3 Boltzmann Distributions and Temperature Dependence 368
 - 7.2.4 Revisiting “What is the Nature of the Activated Complex?” and Why Does TST Work? 369
 - 7.2.5 Experimental Determinations of Activation Parameters and Arrhenius Parameters 370
 - 7.2.6 Examples of Activation Parameters and Their Interpretations 372
 - 7.2.7 Is TST Completely Correct? The Dynamic Behavior of Organic Reactive Intermediates 372

7.3 Postulates and Principles Related to Kinetic Analysis 374

- 7.3.1 The Hammond Postulate 374
- 7.3.2 The Reactivity vs. Selectivity Principle 377
- 7.3.3 The Curtin–Hammett Principle 378
- 7.3.4 Microscopic Reversibility 379
- 7.3.5 Kinetic vs. Thermodynamic Control 380

7.4 Kinetic Experiments	382
7.4.1 How Kinetics Experiments are Performed	382
7.4.2 Kinetic Analyses for Simple Mechanisms	384
<i>First Order Kinetics</i>	385
<i>Second Order Kinetics</i>	386
<i>Pseudo-First Order Kinetics</i>	387
<i>Equilibrium Kinetics</i>	388
<i>Initial-Rate Kinetics</i>	389
<i>Tabulating a Series of Common Kinetic Scenarios</i>	389
7.5 Complex Reactions—Deciphering Mechanisms	390
7.5.1 Steady State Kinetics	390
7.5.2 Using the SSA to Predict Changes in Kinetic Order	395
7.5.3 Saturation Kinetics	396
7.5.4 Prior Rapid Equilibria	397
7.6 Methods for Following Kinetics	397
7.6.1 Reactions with Half-Lives Greater than a Few Seconds	398
7.6.2 Fast Kinetics Techniques	398
<i>Flow Techniques</i>	399
<i>Flash Photolysis</i>	399
<i>Pulse Radiolysis</i>	401
7.6.3 Relaxation Methods	401
7.6.4 Summary of Kinetic Analyses	402
7.7 Calculating Rate Constants	403
7.7.1 Marcus Theory	403
7.7.2 Marcus Theory Applied to Electron Transfer	405
7.8 Considering Multiple Reaction Coordinates	407
7.8.1 Variation in Transition State Structures Across a Series of Related Reactions—An Example Using Substitution Reactions	407
7.8.2 More O’Ferrall–Jencks Plots	409
7.8.3 Changes in Vibrational State Along the Reaction Coordinate—Relating the Third Coordinate to Entropy	412
Summary and Outlook	413
EXERCISES	413
FURTHER READING	417
CHAPTER 8: Experiments Related to Thermodynamics and Kinetics	421
Intent and Purpose	421
8.1 Isotope Effects	421
8.1.1 The Experiment	422
8.1.2 The Origin of Primary Kinetic Isotope Effects	422
<i>Reaction Coordinate Diagrams and Isotope Effects</i>	424
<i>Primary Kinetic Isotope Effects for Linear Transition States as a Function of Exothermicity and Endothermicity</i>	425
<i>Isotope Effects for Linear vs. Non-Linear Transition States</i>	428
8.1.3 The Origin of Secondary Kinetic Isotope Effects	428
<i>Hybridization Changes</i>	429
<i>Steric Isotope Effects</i>	430
8.1.4 Equilibrium Isotope Effects	432
<i>Isotopic Perturbation of Equilibrium—Applications to Carbocations</i>	432
8.1.5 Tunneling	435
8.1.6 Solvent Isotope Effects	437
<i>Fractionation Factors</i>	437
<i>Proton Inventories</i>	438
8.1.7 Heavy Atom Isotope Effects	441
8.1.8 Summary	441
8.2 Substituent Effects	441
8.2.1 The Origin of Substituent Effects	443
<i>Field Effects</i>	443
<i>Inductive Effects</i>	443
<i>Resonance Effects</i>	444
<i>Polarizability Effects</i>	444
<i>Steric Effects</i>	445
<i>Solvation Effects</i>	445
8.3 Hammett Plots—The Most Common LFER. A General Method for Examining Changes in Charges During a Reaction	445
8.3.1 Sigma (σ)	445
8.3.2 Rho (ρ)	447
8.3.3 The Power of Hammett Plots for Deciphering Mechanisms	448
8.3.4 Deviations from Linearity	449
8.3.5 Separating Resonance from Induction	451
8.4 Other Linear Free Energy Relationships	454
8.4.1 Steric and Polar Effects—Taft Parameters	454
8.4.2 Solvent Effects—Grunwald–Winstein Plots	455
8.4.3 Schleyer Adaptation	457
8.4.4 Nucleophilicity and Nucleofugality	458
<i>Basicity/Acidity</i>	459
<i>Solvation</i>	460
<i>Polarizability, Basicity, and Solvation Interplay</i>	460
<i>Shape</i>	461
8.4.5 Swain–Scott Parameters—Nucleophilicity Parameters	461
8.4.6 Edwards and Ritchie Correlations	463
8.5 Acid–Base Related Effects—Brønsted Relationships	464
8.5.1 β_{Nuc}	464
8.5.2 β_{LG}	464
8.5.3 Acid–Base Catalysis	466
8.6 Why do Linear Free Energy Relationships Work?	466
8.6.1 General Mathematics of LFERs	467
8.6.2 Conditions to Create an LFER	468
8.6.3 The Isokinetic or Isoequilibrium Temperature	469
8.6.4 Why does Enthalpy–Entropy Compensation Occur?	469
<i>Steric Effects</i>	470
<i>Solvation</i>	470
8.7 Summary of Linear Free Energy Relationships	470

8.8 Miscellaneous Experiments for Studying Mechanisms 471

- 8.8.1 Product Identification 472
- 8.8.2 Changing the Reactant Structure to Divert or Trap a Proposed Intermediate 473
- 8.8.3 Trapping and Competition Experiments 474
- 8.8.4 Checking for a Common Intermediate 475
- 8.8.5 Cross-Over Experiments 476
- 8.8.6 Stereochemical Analysis 476
- 8.8.7 Isotope Scrambling 477
- 8.8.8 Techniques to Study Radicals: Clocks and Traps 478
- 8.8.9 Direct Isolation and Characterization of an Intermediate 480
- 8.8.10 Transient Spectroscopy 480
- 8.8.11 Stable Media 481

Summary and Outlook 482

EXERCISES 482

FURTHER READING 487

CHAPTER 9: Catalysis 489**Intent and Purpose 489****9.1 General Principles of Catalysis 490**

- 9.1.1 Binding the Transition State *Better* than the Ground State 491
- 9.1.2 A Thermodynamic Cycle Analysis 493
- 9.1.3 A Spatial Temporal Approach 494

9.2 Forms of Catalysis 495

- 9.2.1 "Binding" is Akin to Solvation 495
- 9.2.2 Proximity as a Binding Phenomenon 495
- 9.2.3 Electrophilic Catalysis 499
 - Electrostatic Interactions* 499
 - Metal Ion Catalysis* 500
- 9.2.4 Acid–Base Catalysis 502
- 9.2.5 Nucleophilic Catalysis 502
- 9.2.6 Covalent Catalysis 504
- 9.2.7 Strain and Distortion 505
- 9.2.8 Phase Transfer Catalysis 507

9.3 Brønsted Acid–Base Catalysis 507

- 9.3.1 Specific Catalysis 507
 - The Mathematics of Specific Catalysis* 507
 - Kinetic Plots* 510
- 9.3.2 General Catalysis 510
 - The Mathematics of General Catalysis* 511
 - Kinetic Plots* 512
- 9.3.3 A Kinetic Equivalency 514
- 9.3.4 Concerted or Sequential General-Acid–General-Base Catalysis 515
- 9.3.5 The Brønsted Catalysis Law and Its Ramifications 516
 - A Linear Free Energy Relationship* 516
 - The Meaning of α and β* 517
 - $\alpha + \beta = 1$ 518
 - Deviations from Linearity* 519

9.3.6 Predicting General-Acid or General-Base Catalysis 520

- The Libido Rule* 520
- Potential Energy Surfaces Dictate General or Specific Catalysis* 521

9.3.7 The Dynamics of Proton Transfers 522

- Marcus Analysis* 522

9.4 Enzymatic Catalysis 523

- 9.4.1 Michaelis–Menten Kinetics 523
- 9.4.2 The Meaning of K_M , k_{cat} , and k_{cat}/K_M 524
- 9.4.3 Enzyme Active Sites 525
- 9.4.4 $[S]$ vs. K_M —Reaction Coordinate Diagrams 527
- 9.4.5 Supramolecular Interactions 529

Summary and Outlook 530

EXERCISES 531

FURTHER READING 535

CHAPTER 10: Organic Reaction Mechanisms, Part 1: Reactions Involving Additions and/or Eliminations 537**Intent and Purpose 537****10.1 Predicting Organic Reactivity 538**

- 10.1.1 A Useful Paradigm for Polar Reactions 539
 - Nucleophiles and Electrophiles* 539
 - Lewis Acids and Lewis Bases* 540
 - Donor–Acceptor Orbital Interactions* 540
- 10.1.2 Predicting Radical Reactivity 541
- 10.1.3 In Preparation for the Following Sections 541

—ADDITION REACTIONS— 542**10.2 Hydration of Carbonyl Structures 542**

- 10.2.1 Acid–Base Catalysis 543
- 10.2.2 The Thermodynamics of the Formation of Geminal Diols and Hemiacetals 544

10.3 Electrophilic Addition of Water to Alkenes and Alkynes: Hydration 545

- 10.3.1 Electron Pushing 546
- 10.3.2 Acid-Catalyzed Aqueous Hydration 546
- 10.3.3 Regiochemistry 546
- 10.3.4 Alkyne Hydration 547

10.4 Electrophilic Addition of Hydrogen Halides to Alkenes and Alkynes 548

- 10.4.1 Electron Pushing 548
- 10.4.2 Experimental Observations Related to Regiochemistry and Stereochemistry 548
- 10.4.3 Addition to Alkynes 551

10.5 Electrophilic Addition of Halogens to Alkenes 551

- 10.5.1 Electron Pushing 548
- 10.5.2 Stereochemistry 552

- 10.5.3 Other Evidence Supporting a σ Complex 552
 10.5.4 Mechanistic Variants 553
 10.5.5 Addition to Alkynes 554
- 10.6 Hydroboration** 554
 10.6.1 Electron Pushing 555
 10.6.2 Experimental Observations 555
- 10.7 Epoxidation** 555
 10.7.1 Electron Pushing 556
 10.7.2 Experimental Observations 556
- 10.8 Nucleophilic Additions to Carbonyl Compounds** 556
 10.8.1 Electron Pushing for a Few Nucleophilic Additions 557
 10.8.2 Experimental Observations for Cyanohydrin Formation 559
 10.8.3 Experimental Observations for Grignard Reactions 560
 10.8.4 Experimental Observations in LAH Reductions 561
 10.8.5 Orbital Considerations 561
 The Bürgi–Dunitz Angle 561
 Orbital Mixing 562
 10.8.6 Conformational Effects in Additions to Carbonyl Compounds 562
 10.8.7 Stereochemistry of Nucleophilic Additions 563
- 10.9 Nucleophilic Additions to Olefins** 567
 10.9.1 Electron Pushing 567
 10.9.2 Experimental Observations 567
 10.9.3 Regiochemistry of Addition 567
 10.9.4 Baldwin's Rules 568
- 10.10 Radical Additions to Unsaturated Systems** 569
 10.10.1 Electron Pushing for Radical Additions 569
 10.10.2 Radical Initiators 570
 10.10.3 Chain Transfer vs. Polymerization 571
 10.10.4 Termination 571
 10.10.5 Regiochemistry of Radical Additions 572
- 10.11 Carbene Additions and Insertions** 572
 10.11.1 Electron Pushing for Carbene Reactions 574
 10.11.2 Carbene Generation 574
 10.11.3 Experimental Observations for Carbene Reactions 575
- ELIMINATIONS— 576
- 10.12 Eliminations to Form Carbonyls or "Carbonyl-Like" Intermediates** 577
 10.12.1 Electron Pushing 577
 10.12.2 Stereochemical and Isotope Labeling Evidence 577
 10.12.3 Catalysis of the Hydrolysis of Acetals 578
 10.12.4 Stereoelectronic Effects 579
 10.12.5 CrO_3 Oxidation—The Jones Reagent 580
 Electron Pushing 580
 A Few Experimental Observations 581
- 10.13 Elimination Reactions for Aliphatic Systems—Formation of Alkenes** 581
 10.13.1 Electron Pushing and Definitions 581
 10.13.2 Some Experimental Observations for E2 and E1 Reactions 582
 10.13.3 Contrasting Elimination and Substitution 538
 10.13.4 Another Possibility—E1cB 584
 10.13.5 Kinetics and Experimental Observations for E1cB 584
 10.13.6 Contrasting E2, E1, and E1cB 586
 10.13.7 Regiochemistry of Eliminations 588
 10.13.8 Stereochemistry of Eliminations—Orbital Considerations 590
 10.13.9 Dehydration 592
 Electron Pushing 592
 Other Mechanistic Possibilities 594
 10.13.10 Thermal Eliminations 594
- 10.14 Eliminations from Radical Intermediates** 596
 —COMBINING ADDITION AND ELIMINATION REACTIONS (SUBSTITUTIONS AT sp^2 CENTERS)— 596
- 10.15 The Addition of Nitrogen Nucleophiles to Carbonyl Structures, Followed by Elimination** 597
 10.15.1 Electron Pushing 598
 10.15.2 Acid–Base Catalysis 598
- 10.16 The Addition of Carbon Nucleophiles, Followed by Elimination—The Wittig Reaction** 599
 10.16.1 Electron Pushing 600
- 10.17 Acyl Transfers** 600
 10.17.1 General Electron-Pushing Schemes 600
 10.17.2 Isotope Scrambling 601
 10.17.3 Predicting the Site of Cleavage for Acyl Transfers from Esters 602
 10.17.4 Catalysis 602
- 10.18 Electrophilic Aromatic Substitution** 607
 10.18.1 Electron Pushing for Electrophilic Aromatic Substitutions 607
 10.18.2 Kinetics and Isotope Effects 608
 10.18.3 Intermediate Complexes 608
 10.18.4 Regiochemistry and Relative Rates of Aromatic Substitution 609
- 10.19 Nucleophilic Aromatic Substitution** 611
 10.19.1 Electron Pushing for Nucleophilic Aromatic Substitution 611
 10.19.2 Experimental Observations 611
- 10.20 Reactions Involving Benzyne** 612
 10.20.1 Electron Pushing for Benzyne Reactions 612
 10.20.2 Experimental Observations 613
 10.20.3 Substituent Effects 613
- 10.21 The $\text{S}_{\text{RN}}1$ Reaction on Aromatic Rings** 615
 10.21.1 Electron Pushing 615
 10.21.2 A Few Experimental Observations 615

10.22 Radical Aromatic Substitutions 615

10.22.1 Electron Pushing 615

10.22.2 Isotope Effects 616

10.22.3 Regiochemistry 616

Summary and Outlook 617

EXERCISES 617

FURTHER READING 624

**CHAPTER 11: Organic Reaction Mechanisms,
Part 2: Substitutions at Aliphatic
Centers and Thermal Isomerizations/
Rearrangements 627****Intent and Purpose 627**—SUBSTITUTION α TO A CARBONYL CENTER:
ENOL AND ENOLATE CHEMISTRY— 627**11.1 Tautomerization 628**11.1.1 Electron Pushing for Keto–Enol
Tautomerizations 628

11.1.2 The Thermodynamics of Enol Formation 628

11.1.3 Catalysis of Enolizations 629

11.1.4 Kinetic vs. Thermodynamic Control
in Enolate and Enol Formation 629**11.2 α -Halogenation 631**

11.2.1 Electron Pushing 631

11.2.2 A Few Experimental Observations 631

11.3 α -Alkylations 632

11.3.1 Electron Pushing 632

11.3.2 Stereochemistry: Conformational Effects 633

11.4 The Aldol Reaction 634

11.4.1 Electron Pushing 634

11.4.2 Conformational Effects on the Aldol Reaction 634

—SUBSTITUTIONS ON ALIPHATIC CENTERS— 637

11.5 Nucleophilic Aliphatic Substitution Reactions 63711.5.1 S_N2 and S_N1 Electron-Pushing Examples 637

11.5.2 Kinetics 638

11.5.3 Competition Experiments and Product Analyses 639

11.5.4 Stereochemistry 640

11.5.5 Orbital Considerations 643

11.5.6 Solvent Effects 643

11.5.7 Isotope Effect Data 646

11.5.8 An Overall Picture of S_N2 and S_N1 Reactions 64611.5.9 Structure–Function Correlations
with the Nucleophile 64811.5.10 Structure–Function Correlations
with the Leaving Group 65111.5.11 Structure–Function Correlations
with the R Group 651*Effect of the R Group Structure on S_N2 Reactions* 651*Effect of the R Group Structure on S_N1 Reactions* 653

11.5.12 Carbocation Rearrangements 656

11.5.13 Anchimeric Assistance in S_N1 Reactions 65911.5.14 S_N1 Reactions Involving Non-Classical
Carbocations 661*Norbornyl Cation* 662*Cyclopropyl Carbinyl Carbocation* 66411.5.15 Summary of Carbocation Stabilization
in Various Reactions 66711.5.16 The Interplay Between Substitution
and Elimination 667**11.6 Substitution, Radical, Nucleophilic 668**

11.6.1 The SET Reaction—Electron Pushing 668

11.6.2 The Nature of the Intermediate
in an SET Mechanism 669

11.6.3 Radical Rearrangements as Evidence 669

11.6.4 Structure–Function Correlations
with the Leaving Group 67011.6.5 The $S_{RN}1$ Reaction—Electron Pushing 670**11.7 Radical Aliphatic Substitutions 671**

11.7.1 Electron Pushing 671

11.7.2 Heats of Reaction 671

11.7.3 Regiochemistry of Free Radical
Halogenation 67111.7.4 Autoxidation: Addition of O_2
into C–H Bonds 673*Electron Pushing for Autoxidation* 673

—ISOMERIZATIONS AND REARRANGEMENTS— 674

11.8 Migrations to Electrophilic Carbons 67411.8.1 Electron Pushing for the
Pinacol Rearrangement 67511.8.2 Electron Pushing in the Benzilic Acid
Rearrangement 67511.8.3 Migratory Aptitudes in the Pinacol
Rearrangement 67511.8.4 Stereoelectronic and Stereochemical Considerations
in the Pinacol Rearrangement 67611.8.5 A Few Experimental Observations for the Benzilic
Acid Rearrangement 678**11.9 Migrations to Electrophilic Heteroatoms 678**11.9.1 Electron Pushing in the Beckmann
Rearrangement 67811.9.2 Electron Pushing for the Hofmann
Rearrangement 67811.9.3 Electron Pushing for the Schmidt
Rearrangement 68011.9.4 Electron Pushing for the Baeyer–Villiger
Oxidation 68011.9.5 A Few Experimental Observations for the
Beckmann Rearrangement 68011.9.6 A Few Experimental Observations for the
Schmidt Rearrangement 68111.9.7 A Few Experimental Observations for the
Baeyer–Villiger Oxidation 681**11.10 The Favorskii Rearrangement and Other
Carbanion Rearrangements 682**

11.10.1 Electron Pushing 682

11.10.2 Other Carbanion Rearrangements 683

11.11 Rearrangements Involving Radicals 683

- 11.11.1 Hydrogen Shifts 683
- 11.11.2 Aryl and Vinyl Shifts 684
- 11.11.3 Ring-Opening Reactions 685

11.12 Rearrangements and Isomerizations**Involving Biradicals 685**

- 11.12.1 Electron Pushing Involving Biradicals 685
- 11.12.2 Tetramethylene 687
- 11.12.3 Trimethylene 689
- 11.12.4 Trimethylenemethane 693

Summary and Outlook 695**EXERCISES 695****FURTHER READING 703**

CHAPTER 12: Organotransition Metal Reaction Mechanisms and Catalysis 705
Intent and Purpose 705**12.1 The Basics of Organometallic Complexes 705**

- 12.1.1 Electron Counting and Oxidation State 706
 - Electron Counting* 706
 - Oxidation State* 708
 - d Electron Count* 708
 - Ambiguities* 708
- 12.1.2 The 18-Electron Rule 710
- 12.1.3 Standard Geometries 710
- 12.1.4 Terminology 711
- 12.1.5 Electron Pushing with Organometallic Structures 711
- 12.1.6 *d* Orbital Splitting Patterns 712
- 12.1.7 Stabilizing Reactive Ligands 713

12.2 Common Organometallic Reactions 714

- 12.2.1 Ligand Exchange Reactions 714
 - Reaction Types* 714
 - Kinetics* 716
 - Structure–Function Relationships with the Metal* 716
 - Structure–Function Relationships with the Ligand* 716
 - Substitutions of Other Ligands* 717
- 12.2.2 Oxidative Addition 717
 - Stereochemistry of the Metal Complex* 718
 - Kinetics* 718
 - Stereochemistry of the R Group* 719
 - Structure–Function Relationship for the R Group* 720
 - Structure–Function Relationships for the Ligands* 720
 - Oxidative Addition at sp^2 Centers* 721
 - Summary of the Mechanisms for Oxidative Addition* 721
- 12.2.3 Reductive Elimination 724
 - Structure–Function Relationship for the R Group and the Ligands* 724
 - Stereochemistry at the Metal Center* 725
 - Other Mechanisms* 725
 - Summary of the Mechanisms for Reductive Elimination* 726

12.2.4 α - and β -Eliminations 727

- General Trends for α - and β -Eliminations* 727
- Kinetics* 728
- Stereochemistry of β -Hydride Elimination* 729
- 12.2.5 Migratory Insertions 729
 - Kinetics* 730
 - Studies to Decipher the Mechanism of Migratory Insertion Involving CO* 730
 - Other Stereochemical Considerations* 732
- 12.2.6 Electrophilic Addition to Ligands 733
 - Reaction Types* 733
 - Common Mechanisms Deduced From Stereochemical Analyses* 734
- 12.2.7 Nucleophilic Addition to Ligands 734
 - Reaction Types* 735
 - Stereochemical and Regiochemical Analyses* 735

12.3 Combining the Individual Reactions into Overall Transformations and Cycles 737

- 12.3.1 The Nature of Organometallic Catalysis—Change in Mechanism 738
- 12.3.2 The Monsanto Acetic Acid Synthesis 738
- 12.3.3 Hydroformylation 739
- 12.3.4 The Water–Gas Shift Reaction 740
- 12.3.5 Olefin Oxidation—The Wacker Process 741
- 12.3.6 Palladium Coupling Reactions 742
- 12.3.7 Allylic Alkylation 743
- 12.3.8 Olefin Metathesis 744

Summary and Outlook 747**EXERCISES 748****FURTHER READING 750**

CHAPTER 13: Organic Polymer and Materials Chemistry 753
Intent and Purpose 753**13.1 Structural Issues in Materials Chemistry 754**

- 13.1.1 Molecular Weight Analysis of Polymers 754
 - Number Average and Weight Average Molecular Weights— M_n and M_w* 754
- 13.1.2 Thermal Transitions—Thermoplastics and Elastomers 757
- 13.1.3 Basic Polymer Topologies 759
- 13.1.4 Polymer–Polymer Phase Behavior 760
- 13.1.5 Polymer Processing 762
- 13.1.6 Novel Topologies—Dendrimers and Hyperbranched Polymers 763
 - Dendrimers* 763
 - Hyperbranched Polymers* 768
- 13.1.7 Liquid Crystals 769
- 13.1.8 Fullerenes and Carbon Nanotubes 775

13.2 Common Polymerization Mechanisms 779

- 13.2.1 General Issues 779
- 13.2.2 Polymerization Kinetics 782
 - Step-Growth Kinetics* 782

<i>Free-Radical Chain Polymerization</i>	783
<i>Living Polymerizations</i>	785
<i>Thermodynamics of Polymerizations</i>	787
13.2.3 Condensation Polymerization	788
13.2.4 Radical Polymerization	791
13.2.5 Anionic Polymerization	793
13.2.6 Cationic Polymerization	794
13.2.7 Ziegler–Natta and Related Polymerizations	794
<i>Single-Site Catalysts</i>	796
13.2.8 Ring-Opening Polymerization	797
13.2.9 Group Transfer Polymerization (GTP)	799

Summary and Outlook 800

EXERCISES 801

FURTHER READING 803

PART III

ELECTRONIC STRUCTURE: THEORY AND APPLICATIONS

CHAPTER 14: Advanced Concepts in Electronic Structure Theory 807

Intent and Purpose 807

14.1 Introductory Quantum Mechanics 808

- 14.1.1 The Nature of Wavefunctions 808
- 14.1.2 The Schrödinger Equation 809
- 14.1.3 The Hamiltonian 809
- 14.1.4 The Nature of the ∇^2 Operator 811
- 14.1.5 Why Do Bonds Form? 812

14.2 Computational Methods—Solving the Schrödinger Equation for Complex Systems 815

- 14.2.1 *Ab Initio* Molecular Orbital Theory 815
 - Born–Oppenheimer Approximation* 815
 - The Orbital Approximation* 815
 - Spin* 816
 - The Pauli Principle and Determinantal Wave Functions* 816
 - The Hartree–Fock Equation and the Variational Theorem* 818
 - SCF Theory* 821
 - Linear Combination of Atomic Orbitals—Molecular Orbitals (LCAO–MO)* 821
 - Common Basis Sets—Modeling Atomic Orbitals* 822
 - Extension Beyond HF—Correlation Energy* 824
 - Solvation* 825
 - General Considerations* 825
 - Summary* 826
- 14.2.2 Secular Determinants—A Bridge Between *Ab Initio*, Semi-Empirical/ Approximate, and Perturbational Molecular Orbital Theory Methods 828
 - The “Two-Orbital Mixing Problem”* 829

Writing the Secular Equations and Determinant for Any Molecule 832

- 14.2.3 Semi-Empirical and Approximate Methods 833
 - Neglect of Differential Overlap (NDO) Methods* 833
 - i. CNDO, INDO, PNDO (C = Complete, I = Intermediate, P = Partial)* 834
 - ii. The Semi-Empirical Methods: MNDO, AM1, and PM3* 834
 - Extended Hückel Theory (EHT)* 834
 - Hückel Molecular Orbital Theory (HMOT)* 835
- 14.2.4 Some General Comments on Computational Quantum Mechanics 835
- 14.2.5 An Alternative: Density Functional Theory (DFT) 836

14.3 A Brief Overview of the Implementation and Results of HMOT 837

- 14.3.1 Implementing Hückel Theory 838
- 14.3.2 HMOT of Cyclic π Systems 840
- 14.3.3 HMOT of Linear π Systems 841
- 14.3.4 Alternate Hydrocarbons 842

14.4 Perturbation Theory—Orbital Mixing Rules 844

- 14.4.1 Mixing of Degenerate Orbitals—First-Order Perturbations 845
- 14.4.2 Mixing of Non-Degenerate Orbitals—Second-Order Perturbations 845

14.5 Some Topics in Organic Chemistry for Which Molecular Orbital Theory Lends Important Insights 846

- 14.5.1 Arenes: Aromaticity and Antiaromaticity 846
- 14.5.2 Cyclopropane and Cyclopropylcarbinyl—Walsh Orbitals 848
 - The Cyclic Three-Orbital Mixing Problem* 849
 - The MOs of Cyclopropane* 850
- 14.5.3 Planar Methane 853
- 14.5.4 Through-Bond Coupling 854
- 14.5.5 Unique Bonding Capabilities of Carbocations—Non-Classical Ions and Hypervalent Carbon 855
 - Transition State Structure Calculations* 856
 - Application of These Methods to Carbocations* 857
 - NMR Effects in Carbocations* 857
 - The Norbornyl Cation* 858
- 14.5.6 Spin Preferences 859
 - Two Weakly Interacting Electrons: H₂ vs. Atomic C* 859

14.6 Organometallic Complexes 862

- 14.6.1 Group Orbitals for Metals 863
- 14.6.2 The Isolobal Analogy 866
- 14.6.3 Using the Group Orbitals to Construct Organometallic Complexes 867

Summary and Outlook 868

EXERCISES 868

FURTHER READING 875

CHAPTER 15: Thermal Pericyclic Reactions 877**Intent and Purpose** 877**15.1 Background** 878**15.2 A Detailed Analysis of Two Simple Cycloadditions** 878

- 15.2.1 Orbital Symmetry Diagrams 879
 [2+2] 879
 [4+2] 881

- 15.2.2 State Correlation Diagrams 883
 [2+2] 883
 [4+2] 886

- 15.2.3 Frontier Molecular Orbital (FMO) Theory 888
 Contrasting the [2+2] and [4+2] 888

- 15.2.4 Aromatic Transition State Theory/Topology 889

- 15.2.5 The Generalized Orbital Symmetry Rule 890

- 15.2.6 Some Comments on “Forbidden” and “Allowed” Reactions 892

- 15.2.7 Photochemical Pericyclic Reactions 892

- 15.2.8 Summary of the Various Methods 893

15.3 Cycloadditions 893

- 15.3.1 An Allowed Geometry for [2+2] Cycloadditions 894

- 15.3.2 Summarizing Cycloadditions 895

- 15.3.3 General Experimental Observations 895

- 15.3.4 Stereochemistry and Regiochemistry of the Diels–Alder Reaction 896
 An Orbital Approach to Predicting Regiochemistry 896
 The Endo Effect 899

- 15.3.5 Experimental Observations for [2+2] Cycloadditions 901

- 15.3.6 Experimental Observations for 1,3-Dipolar Cycloadditions 901

- 15.3.7 Retrocycloadditions 902

15.4 Electrocyclic Reactions 903

- 15.4.1 Terminology 903

- 15.4.2 Theoretical Analyses 904

- 15.4.3 Experimental Observations: Stereochemistry 906

- 15.4.4 Torquoselectivity 908

15.5 Sigmatropic Rearrangements 910

- 15.5.1 Theory 911

- 15.5.2 Experimental Observations: A Focus on Stereochemistry 913

- 15.5.3 The Mechanism of the Cope Rearrangement 916

- 15.5.4 The Claisen Rearrangement 921
 Uses in Synthesis 921
 Mechanistic Studies 923

- 15.5.5 The Ene Reaction 924

15.6 Cheletropic Reactions 924

- 15.6.1 Theoretical Analyses 926

- 15.6.2 Carbene Additions 927

15.7 In Summary—Applying the Rules 928**Summary and Outlook** 928**EXERCISES** 929**FURTHER READING** 933

CHAPTER 16: Photochemistry 935**Intent and Purpose** 935**16.1 Photophysical Processes—The Jablonski Diagram** 936

- 16.1.1 Electromagnetic Radiation 936

Multiple Energy Surfaces Exist 937

- 16.1.2 Absorption 939

- 16.1.3 Radiationless Vibrational Relaxation 944

- 16.1.4 Fluorescence 945

- 16.1.5 Internal Conversion (IC) 949

- 16.1.6 Intersystem Crossing (ISC) 950

- 16.1.7 Phosphorescence 951

- 16.1.8 Quantum Yield 952

- 16.1.9 Summary of Photophysical Processes 952

16.2 Bimolecular Photophysical Processes 953

- 16.2.1 General Considerations 953

- 16.2.2 Quenching, Excimers, and Exciplexes 953
 Quenching 954
 Excimers and Exciplexes 954

Photoinduced Electron Transfer 955

- 16.2.3 Energy Transfer I. The Dexter Mechanism—Sensitization 956

- 16.2.4 Energy Transfer II. The Förster Mechanism 958

- 16.2.5 FRET 960

- 16.2.6 Energy Pooling 962

- 16.2.7 An Overview of Bimolecular Photophysical Processes 962

16.3 Photochemical Reactions 962

- 16.3.1 Theoretical Considerations—Funnels 962
 Adiabatic Photoreactions 963
 Other Mechanisms 964

- 16.3.2 Acid–Base Chemistry 965

- 16.3.3 Olefin Isomerization 965

- 16.3.4 Reversal of Pericyclic Selection Rules 968

- 16.3.5 Photocycloaddition Reactions 970
 Making Highly Strained Ring Systems 973
 Breaking Aromaticity 974

- 16.3.6 The Di- π -Methane Rearrangement 974

- 16.3.7 Carbonyls Part I: The Norrish I Reaction 976

- 16.3.8 Carbonyls Part II: Photoreduction and the Norrish II Reaction 978

- 16.3.9 Nitrobenzyl Photochemistry: “Caged” Compounds 980

- 16.3.10 Elimination of N₂: Azo Compounds, Diazo Compounds, Diazirines, and Azides 981
 - Azoalkanes (1,2-Diazenes)* 981
 - Diazo Compounds and Diazirines* 982
 - Azides* 983

16.4 Chemiluminescence 985

- 16.4.1 Potential Energy Surface for a Chemiluminescent Reaction 985
- 16.4.2 Typical Chemiluminescent Reactions 986
- 16.4.3 Dioxetane Thermolysis 987

16.5 Singlet Oxygen 989

Summary and Outlook 993

EXERCISES 993

FURTHER READING 999

CHAPTER 17: Electronic Organic Materials 1001

Intent and Purpose 1001

17.1 Theory 1001

- 17.1.1 Infinite π Systems—An Introduction to Band Structures 1002
- 17.1.2 The Peierls Distortion 1009
- 17.1.3 Doping 1011

17.2 Conducting Polymers 1016

- 17.2.1 Conductivity 1016
- 17.2.2 Polyacetylene 1017
- 17.2.3 Polyarenes and Polyarenevinylenes 1018
- 17.2.4 Polyaniline 1021

17.3 Organic Magnetic Materials 1022

- 17.3.1 Magnetism 1023
- 17.3.2 The Molecular Approach to Organic Magnetic Materials 1024
- 17.3.3 The Polymer Approach to Organic Magnetic Materials—Very High-Spin Organic Molecules 1027

17.4 Superconductivity 1030

- 17.4.1 Organic Metals/Synthetic Metals 1032

17.5 Non-Linear Optics (NLO) 1033

17.6 Photoresists 1036

- 17.6.1 Photolithography 1036
- 17.6.2 Negative Photoresists 1037
- 17.6.3 Positive Photoresists 1038

Summary and Outlook 1041

EXERCISES 1042

FURTHER READING 1044

APPENDIX 1: Conversion Factors and Other Useful Data 1047

APPENDIX 2: Electrostatic Potential Surfaces for Representative Organic Molecules 1049

APPENDIX 3: Group Orbitals of Common Functional Groups: Representative Examples Using Simple Molecules 1051

APPENDIX 4: The Organic Structures of Biology 1057

APPENDIX 5: Pushing Electrons 1061

- A5.1 The Rudiments of Pushing Electrons 1061
- A5.2 Electron Sources and Sinks for Two-Electron Flow 1062
- A5.3 How to Denote Resonance 1064
- A5.4 Common Electron-Pushing Errors 1065
 - Backwards Arrow Pushing 1065
 - Not Enough Arrows 1065
 - Losing Track of the Octet Rule 1066
 - Losing Track of Hydrogens and Lone Pairs 1066
 - Not Using the Proper Source 1067
 - Mixed Media Mistakes 1067
 - Too Many Arrows—Short Cuts 1067
- A5.5 Complex Reactions—Drawing a Chemically Reasonable Mechanism 1068
- A5.6 Two Case Studies of Predicting Reaction Mechanisms 1069
- A5.7 Pushing Electrons for Radical Reactions 1071
 - Practice Problems for Pushing Electrons 1073

APPENDIX 6: Reaction Mechanism Nomenclature 1075

Index 1079

List of Highlights

CHAPTER 1

- How Realistic are Formal Charges? 7
- NMR Coupling Constants 10
- Scaling Electrostatic Surface Potentials 15
- 1-Fluorobutane 16
- Particle in a Box 21
- Resonance in the Peptide Amide Bond? 23
- A Brief Look at Symmetry and Symmetry Operations 29
- CH_5^+ —Not Really a Well-Defined Structure 55
- Pyramidal Inversion: NH_3 vs. PH_3 57
- Stable Carbenes 59

CHAPTER 2

- Entropy Changes During Cyclization Reactions 71
- A Consequence of High Bond Strength:
The Hydroxyl Radical in Biology 73
- The Half-Life for Homolysis of Ethane
at Room Temperature 73
- The Probability of Finding Atoms at Particular
Separations 75
- How Do We Know That $n = 0$ is Most Relevant
for Bond Stretches at $T = 298 \text{ K}$? 76
- Potential Surfaces for Bond Bending Motions 78
- How Big is 3 kcal/mol ? 93
- Shouldn't Torsional Motions be Quantized? 94
- The Geometry of Radicals 96
- Differing Magnitudes of Energy Values in Thermodynamics
and Kinetics 100
- "Sugar Pucker" in Nucleic Acids 102
- Alternative Measurements of Steric Size 104
- The Use of A Values in a Conformational Analysis
Study for the Determination of Intramolecular
Hydrogen Bond Strength 105
- The NMR Time Scale 106
- Ring Fusion—Steroids 108
- A Conformational Effect on the Material Properties
of Poly(3-Alkylthiophenes) 116
- Cyclopropenyl Cation 117
- Cyclopropenyl Anion 118
- Porphyrins 119
- Protein Disulfide Linkages 123
- From Strained Molecules to Molecular Rods 126
- Cubane Explosives? 126
- Molecular Gears 128

CHAPTER 3

- The Use of Solvent Scales to Direct Diels–Alder
Reactions 149
- The Use of Wetting and the Capillary Action
Force to Drive the Self-Assembly of
Macroscopic Objects 151
- The Solvent Packing Coefficient and
the 55% Solution 152
- Solvation Can Affect Equilibria 155
- A van't Hoff Analysis of the Formation of a
Stable Carbene 163

- The Strength of a Buried Salt Bridge 165
- The Angular Dependence of Dipole–Dipole Interactions—
The "Magic Angle" 168
- An Unusual Hydrogen Bond Acceptor 169
- Evidence for Weak Directionality Considerations 170
- Intramolecular Hydrogen Bonds are Best
for Nine-Membered Rings 170
- Solvent Scales and Hydrogen Bonds 172
- The Extent of Resonance can be Correlated with
Hydrogen Bond Length 174
- Cooperative Hydrogen Bonding in Saccharides 175
- How Much is a Hydrogen Bond in an α -Helix Worth? 176
- Proton Sponges 179
- The Relevance of Low-Barrier Hydrogen Bonds
to Enzymatic Catalysis 179
- β -Peptide Foldamers 180
- A Cation– π Interaction at the Nicotine Receptor 183
- The Polar Nature of Benzene Affects Acidities
in a Predictable Manner 184
- Use of the Arene–Perfluorarene Interaction in the
Design of Solid State Structures 185
- Donor–Acceptor Driven Folding 187
- The Hydrophobic Effect and Protein Folding 194
- More Foldamers: Folding Driven by
Solvophobic Effects 195
- Calculating Drug Binding Energies by SPT 201

CHAPTER 4

- The Units of Binding Constants 209
- Cooperativity in Drug Receptor Interactions 215
- The Hill Equation and Cooperativity in
Protein–Ligand Interactions 219
- The Benesi–Hildebrand Plot 221
- How are Heat Changes Related to Enthalpy? 223
- Using the Helical Structure of Peptides and the
Complexation Power of Crowns to Create
an Artificial Transmembrane Channel 226
- Preorganization and the Salt Bridge 229
- A Clear Case of Entropy Driven Electrostatic
Complexation 229
- Salt Bridges Evaluated by Non-Biological Systems 230
- Does Hydrogen Bonding *Really* Play a Role in
DNA Strand Recognition? 233
- Calixarenes—Important Building Blocks for Molecular
Recognition and Supramolecular Chemistry 238
- Aromatics at Biological Binding Sites 239
- Combining the Cation– π Effect and Crown Ethers 240
- A Thermodynamic Cycle to Determine the Strength
of a Polar– π Interaction 242
- Molecular Mechanics/Modeling and Molecular
Recognition 243
- Biotin/ Avidin: A Molecular Recognition/
Self-Assembly Tool from Nature 249
- Taming Cyclobutadiene—A Remarkable Use of
Supramolecular Chemistry 251

CHAPTER 5

- Using a pH Indicator to Sense Species Other Than the Hydronium Ion 264
- Realistic Titrations in Water 265
- An Extremely Acidic Medium is Formed During Photo-Initiated Cationic Polymerization in Photolithography 269
- Super Acids Used to Activate Hydrocarbons 270
- The Intrinsic Acidity Increase of a Carbon Acid by Coordination of BF_3 276
- Direct Observation of Cytosine Protonation During Triple Helix Formation 287
- A Shift of the Acidity of an N–H Bond in Water Due to the Proximity of an Ammonium or Metal Cation 288
- The Notion of Super electrophiles Produced by Super Acids 289

CHAPTER 6

- Stereoisomerism and Connectivity 300
- Total Synthesis of an Antibiotic with a Staggering Number of Stereocenters 303
- The Descriptors for the Amino Acids Can Lead to Confusion 307
- Chiral Shift Reagents 308
- C_2 Ligands in Asymmetric Synthesis 313
- Enzymatic Reactions, Molecular Imprints, and Enantiotopic Discrimination 320
- Biological Knots—DNA and Proteins 325
- Polypropylene Structure and the Mass of the Universe 331
- Controlling Polymer Tacticity—The Metallocenes 332
- CD Used to Distinguish α -Helices from β -Sheets 335
- Creating Chiral Phosphates for Use as Mechanistic Probes 335
- A Molecular Helix Created from Highly Twisted Building Blocks 338

CHAPTER 7

- Single-Molecule Kinetics 360
- Using the Arrhenius Equation to Determine Differences in Activation Parameters for Two Competing Pathways 371
- Curvature in an Eyring Plot is Used as Evidence for an Enzyme Conformational Change in the Catalysis of the Cleavage of the Co–C Bond of Vitamin B_{12} 371
- Where TST May be Insufficient 374
- The Transition States for $\text{S}_{\text{N}}1$ Reactions 377
- Comparing Reactivity to Selectivity in Free Radical Halogenation 378
- Using the Curtin–Hammett Principle to Predict the Stereochemistry of an Addition Reaction 379
- Applying the Principle of Microscopic Reversibility to Phosphate Ester Chemistry 380
- Kinetic vs. Thermodynamic Enolates 382
- Molecularity vs. Mechanism. Cyclization Reactions and Effective Molarity 384
- First Order Kinetics: Delineating Between a Unimolecular and a Bimolecular Reaction of Cyclopentyne and Dienes 386
- The Observation of Second Order Kinetics to Support a Multistep Displacement Mechanism for a Vitamin Analog 387

- Pseudo-First Order Kinetics: Revisiting the Cyclopentyne Example 388
- Zero Order Kinetics 393
- An Organometallic Example of Using the SSA to Delineate Mechanisms 395
- Saturation Kinetics That We Take for Granted— $\text{S}_{\text{N}}1$ Reactions 397
- Prior Equilibrium in an $\text{S}_{\text{N}}1$ Reaction 398
- Femtochemistry: Direct Characterization of Transition States 400
- “Seeing” Transition States Part II: The Role of Computation 401
- The Use of Pulse Radiolysis to Measure the pK_{a} s of Protonated Ketyl Anions 402
- Discovery of the Marcus Inverted Region 406
- Using a More O’Ferrall–Jencks Plot in Catalysis 400

CHAPTER 8

- The Use of Primary Kinetic Isotope Effects to Probe the Mechanism of Aliphatic Hydroxylation by Iron(III) Porphyrins 425
- An Example of Changes in the Isotope Effect with Varying Reaction Free Energies 428
- The Use of an Inverse Isotope Effect to Delineate an Enzyme Mechanism 431
- An Ingenious Method for Measuring Very Small Isotope Effects 432
- An Example of Tunneling in a Common Synthetic Organic Reaction 436
- Using Fractionation Factors to Characterize Very Strong Hydrogen Bonds 439
- The Use of a Proton Inventory to Explore the Mechanism of Ribonuclease Catalysis 440
- A Substituent Effect Study to Decipher the Reason for the High Stability of Collagen 444
- Using a Hammett Plot to Explore the Behavior of a Catalytic Antibody 450
- An Example of a Change in Mechanism in a Solvolysis Reaction Studied Using σ^+ 452
- A Swain–Lupton Correlation for Tungsten-Bipyridine-Catalyzed Allylic Alkylation 453
- Using Taft Parameters to Understand the Structures of Cobaloximes; Vitamin B_{12} Mimics 455
- The Use of the Schleyer Method to Determine the Extent of Nucleophilic Assistance in the Solvolysis of Arylvinyl Tosylates 459
- The Use of Swain–Scott Parameters to Determine the Mechanism of Some Acetal Substitution Reactions 459
- ATP Hydrolysis—How β_{LG} and β_{Nuc} Values Have Given Insight into Transition State Structures 465
- How Can Some Groups be Both Good Nucleophiles and Good Leaving Groups? 466
- An Example of an Unexpected Product 472
- Designing a Method to Divert the Intermediate 473
- Trapping a Phosphorane Legitimizes its Existence 474
- Checking for a Common Intermediate in Rhodium-Catalyzed Allylic Alkylations 475
- Pyranoside Hydrolysis by Lysozyme 476
- Using Isotopic Scrambling to Distinguish Exocyclic vs. Endocyclic Cleavage Pathways for a Pyranoside 478

Determination of 1,4-Biradical Lifetimes Using
a Radical Clock 480
The Identification of Intermediates from a Catalytic Cycle
Needs to be Interpreted with Care 481

CHAPTER 9

The Application of Figure 9.4 to Enzymes 494
High Proximity Leads to the Isolation of a Tetrahedral
Intermediate 498
The Notion of “Near Attack Conformations” 499
Toward an Artificial Acetylcholinesterase 501
Metal and Hydrogen Bonding Promoted Hydrolysis
of 2',3'-cAMP 502
Nucleophilic Catalysis of Electrophilic Reactions 503
Organocatalysis 505
Lysozyme 506
A Model for General-Acid–General-Base Catalysis 514
Anomalous Brønsted Values 519
Artificial Enzymes: Cyclodextrins Lead the Way 530

CHAPTER 10

Cyclic Forms of Saccharides and Concerted Proton
Transfers 545
Squalene to Lanosterol 550
Mechanisms of Asymmetric Epoxidation Reactions 558
Nature's Hydride Reducing Agent 566
The Captodative Effect 573
Stereoelectronics in an Acyl Transfer Model 579
The Swern Oxidation 580
Gas Phase Eliminations 588
Using the Curtin–Hammett Principle 593
Aconitase—An Enzyme that Catalyzes Dehydration
and Rehydration 595
Enzymatic Acyl Transfers I: The Catalytic Triad 604
Enzymatic Acyl Transfers II: Zn(II) Catalysis 605
Enzyme Mimics for Acyl Transfers 606
Peptide Synthesis—Optimizing Acyl Transfer 606

CHAPTER 11

Enolate Aggregation 631
Control of Stereochemistry in Enolate Reactions 636
Gas Phase S_N2 Reactions—A Stark Difference in Mechanism
from Solution 641
A Potential Kinetic Quandary 642
Contact Ion Pairs vs. Solvent-Separated Ion Pairs 647
An Enzymatic S_N2 Reaction: Haloalkane
Dehydrogenase 649
The Meaning of β_{LG} Values 651
Carbocation Rearrangements in Rings 658
Anchimeric Assistance in War 660
Further Examples of Hypervalent Carbon 666
Brominations Using *N*-Bromosuccinimide 673
An Enzymatic Analog to the Benzilic Acid Rearrangement:
Acetohydroxy-Acid Isomeroreductase 677

CHAPTER 12

Bonding Models 709
Electrophilic Aliphatic Substitutions (S_E2 and S_E1) 715
C–H Activation, Part 1 722
C–H Activation, Part 2 723

The Sandmeyer Reaction 726
Olefin Slippage During Nucleophilic Addition to
Alkenes 737
Pd(0) Coupling Reactions in Organic Synthesis 742
Stereocontrol at Every Step in Asymmetric Allylic
Alkylations 745
Cyclic Rings Possessing Over 100,000 Carbons! 747

CHAPTER 13

Monodisperse Materials Prepared Biosynthetically 756
An Analysis of Dispersity and Molecular Weight 757
A Melting Analysis 759
Protein Folding Modeled by a Two-State Polymer
Phase Transition 762
Dendrimers, Fractals, Neurons, and Trees 769
Lyotropic Liquid Crystals: From Soap Scum to
Biological Membranes 774
Organic Surfaces: Self-Assembling Monolayers and
Langmuir–Blodgett Films 778
Free-Radical Living Polymerizations 787
Lycra/Spandex 790
Radical Copolymerization—Not as Random
as You Might Think 792
PMMA—One Polymer with a Remarkable Range
of Uses 793
Living Polymers for Better Running Shoes 795
Using ^{13}C NMR Spectroscopy to Evaluate Polymer
Stereochemistry 797

CHAPTER 14

The Hydrogen Atom 811
Methane—Molecular Orbitals or Discrete Single
Bonds with sp^3 Hybrids? 827
Koopmans' Theorem—A Connection Between *Ab Initio*
Calculations and Experiment 828
A Matrix Approach to Setting Up the LCAO Method 832
Through-Bond Coupling and Spin Preferences 861
Cyclobutadiene at the Two-Electron Level of Theory 862

CHAPTER 15

Symmetry Does Matter 887
Allowed Organometallic [2+2] Cycloadditions 895
Semi-Empirical vs. *Ab Initio* Treatments of Pericyclic
Transition States 900
Electrocyclization in Cancer Therapeutics 910
Fluxional Molecules 913
A Remarkable Substituent Effect: The Oxy-Cope
Rearrangement 921
A Biological Claisen Rearrangement—The Chorismate
Mutase Reaction 922
Hydrophobic Effects in Pericyclic Reactions 923
Pericyclic Reactions of Radical Cations 925

CHAPTER 16

Excited State Wavefunctions 937
Physical Properties of Excited States 944
The Sensitivity of Fluorescence—Good News and
Bad News 946
GFP Part I: Nature's Fluorophore 947
Isosbestic Points—Hallmarks of One-to-One Stoichiometric
Conversions 949

The “Free Rotor” or “Loose Bolt” Effect on
Quantum Yields 953
Single-Molecule FRET 961
Trans-Cyclohexene? 967
Retinal and Rhodopsin—The Photochemistry
of Vision 968
Photochromism 969
UV Damage of DNA—A [2+2] Photoreaction 971
Using Photochemistry to Generate Reactive Intermediates:
Strategies Fast and Slow 983

Photoaffinity Labeling—A Powerful Tool for
Chemical Biology 984
Light Sticks 987
GFP Part II: Aequorin 989
Photodynamic Therapy 991

CHAPTER 17

Solitons in Polyacetylene 1015
Scanning Probe Microscopy 1040
Soft Lithography 1041

The twentieth century saw the birth of physical organic chemistry—the study of the interrelationships between structure and reactivity in organic molecules—and the discipline matured to a brilliant and vibrant field. Some would argue that the last century also saw the near death of the field. Undeniably, physical organic chemistry has had some difficult times. There is a perception by some that chemists thoroughly understand organic reactivity and that there are no *important* problems left. This view ignores the fact that while the rigorous treatment of structure and reactivity in organic structures that is the field's hallmark continues, physical organic chemistry has expanded to encompass other disciplines.

In our opinion, physical organic chemistry is alive and well in the early twenty-first century. New life has been breathed into the field because it has embraced newer chemical disciplines, such as bioorganic, organometallic, materials, and supramolecular chemistries. Bioorganic chemistry is, to a considerable extent, physical organic chemistry on proteins, nucleic acids, oligosaccharides, and other biomolecules. Organometallic chemistry traces its intellectual roots directly to physical organic chemistry, and the tools and conceptual framework of physical organic chemistry continue to permeate the field. Similarly, studies of polymers and other materials challenge chemists with problems that benefit directly from the techniques of physical organic chemistry. Finally, advances in supramolecular chemistry result from a deeper understanding of the physical organic chemistry of intermolecular interactions. These newer disciplines have given physical organic chemists fertile ground in which to study the interrelationships of structure and reactivity. Yet, even while these new fields have been developing, remarkable advances in our understanding of basic organic chemical reactivity have continued to appear, exploiting classical physical organic tools and developing newer experimental and computational techniques. These new techniques have allowed the investigation of reaction mechanisms with amazing time resolution, the direct characterization of classically elusive molecules such as cyclobutadiene, and highly detailed and accurate computational evaluation of problems in reactivity. Importantly, the techniques of physical organic chemistry and the intellectual approach to problems embodied by the discipline remain as relevant as ever to organic chemistry. Therefore, a course in physical organic chemistry will be essential for students for the foreseeable future.

This book is meant to capture the state of the art of physical organic chemistry in the early twenty-first century, and, within the best of our ability, to present material that will remain relevant as the field evolves in the future. For some time it has been true that if a student opens a physical organic chemistry textbook to a random page, the odds are good that he or she will see very interesting chemistry, but chemistry that does not represent an area of significant current research activity. We seek to rectify that situation with this text. A student must know the fundamentals, such as the essence of structure and bonding in organic molecules, the nature of the basic reactive intermediates, and organic reaction mechanisms. However, students should also have an appreciation of the current issues and challenges in the field, so that when they inspect the modern literature they will have the necessary background to read and understand current research efforts. Therefore, while treating the fundamentals, we have wherever possible chosen examples and highlights from modern research areas. Further, we have incorporated chapters focused upon several of the modern disciplines that benefit from a physical organic approach. From our perspective, a protein, electrically conductive polymer, or organometallic complex should be as relevant to a course in physical organic chemistry as are small rings, annulenes, or nonclassical ions.

We recognize that this is a delicate balancing act. A course in physical organic chemistry

cannot also be a course in bioorganic or materials chemistry. However, a physical organic chemistry class should not be a history course, either. We envision this text as appropriate for many different kinds of courses, depending on which topics the instructor chooses to emphasize. In addition, we hope the book will be the first source a researcher approaches when confronted with a new term or concept in the primary literature, and that the text will provide a valuable introduction to the topic. Ultimately, we hope to have produced a text that will provide the fundamental principles and techniques of physical organic chemistry, while also instilling a sense of excitement about the varied research areas impacted by this brilliant and vibrant field.

Eric V. Anslyn

Norman Hackerman Professor
University Distinguished Teaching Professor
University of Texas, Austin

Dennis A. Dougherty

George Grant Hoag Professor of Chemistry
California Institute of Technology

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they will all be incorporated, but this literature will help us to keep a broad perspective on where the field is moving.

Given the magnitude and scope of this project, we are sure that some unclear presentations, misrepresentations, and even outright errors have crept in. We welcome corrections and comments on these issues from our colleagues around the world. Many difficult choices had to be made over the six years it took to create this text, and no doubt the selection of topics is biased by our own perceptions and interests. We apologize in advance to any of our colleagues who feel their work is not properly represented, and again welcome suggestions.

We wish you the best of luck in using this textbook.

Modern Physical Organic Chemistry
